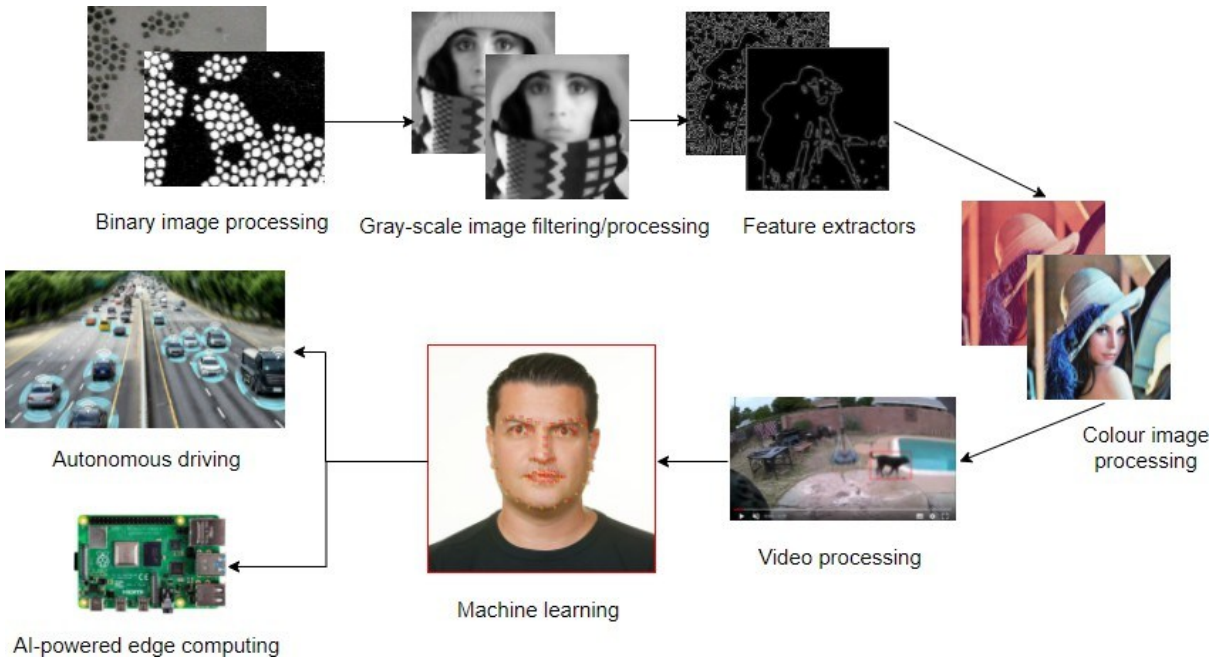
Fundamentals of Object Detection

In this module, you learnt one of the common applications of deep learning in computer vision, that is, **object detection**.

Computer vision (CV) allows a computer **to perceive** and **perform operations** or **tasks** on digital images

CV began in the 1960s as a simple collection of methods for processing black and white images produced by early industrial cameras. Over time, it has evolved to process greyscale and colour images (both 2D and 3D) as well as videos, utilising increasingly complex algorithms. Now, CV applications can be used for complex tasks, such as self-driving cars and face detection. The image given below shows the evolution of CV's applications.

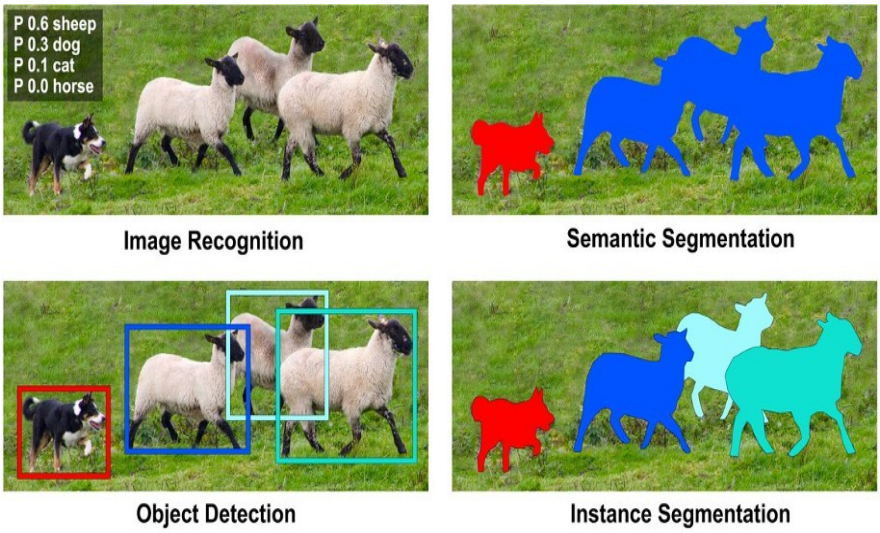


Modern CV, in combination with machine learning (ML), is often tasked with addressing four common problems, which are:

- Image classification,
- Object detection,
- Semantic segmentation and
- Instance segmentation.

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The image given below shows the applications of CV in addressing these four problems.



Introduction to Object Detection

You started this session by learning the key differences between object detection and image classification. **Image classification** algorithms will be able to categorise different objects inside an image into different classes. However, **object detection** algorithms would not only be able to classify the objects but also provide you with their locations in the image. Let's take the following image as an example to understand the difference between these algorithms better.



Let's try and understand the difference between the outputs of the two algorithms in the context of this image:

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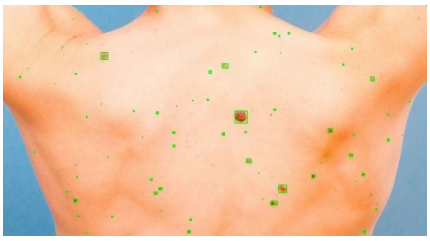
1. Image classification algorithms analyse images and the objects present inside them and then divide them into different classes:
    a. A binary classifier, as the name suggests, categorises the objects in this image into two classes, such as vehicle or not, and so on.
    b. A multiclass classifier, on the other hand, categorises objects into multiple classes and tells whether the object present is a car, a bus, a bike, etc.

2. The output in the case of object detection algorithm will include the above-mentioned classes along with the locations of the car, the bus and the bike, as shown in the image below.



Some of the common applications of object detection algorithms are mentioned below:

1. **Medical imaging:**



2. **Remote sensing:**

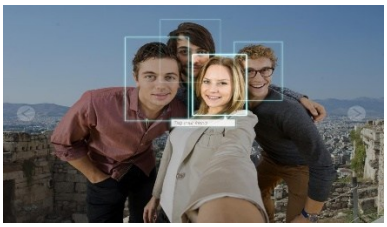


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3. **Self-driving cars:**



4. **Face recognition:**



5. **Traffic monitoring:**



The coloured boxes that you see around the objects in this image are called **bounding boxes**. These boxes provide the coordinates of the objects detected inside an image.

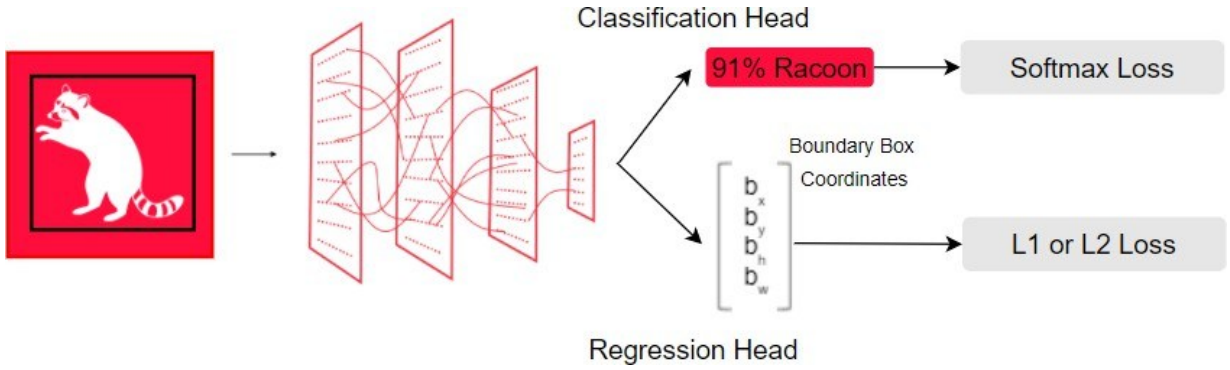
Object Localisation

An object detection algorithm performs two tasks, which are:

a. Detecting an object's class and
b. Identifying the object's location through 'bounding boxes'.

The following image provides a summary of the working of object detection algorithms.

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Object localisation refers to determining the locations of the objects present in an image. The location of the objects is given by the coordinates of the **bounding boxes**.

A **bounding box vector** consists of four elements, which are:

- The x coordinate of the centroid,
- The y coordinate of the centroid,
- The height of the bounding box and
- The width of the bounding box.

In vector form, a bounding box is represented as shown below.

$$\begin{bmatrix} b_x \\ b_y \\ b_h \\ b_w \end{bmatrix}$$

Now, since the output of an object detection model is primarily a bounding box, you need a method to be able to evaluate the efficiency of a model to localise an object and predict its bounding box accurately. The standard industry metric that you need to consider for this purpose is **intersection over union** (IoU).

This metric evaluates the accuracy of your model's prediction of the bounding box by comparing it with the '**ground truth**' bounding box present in the training set. Take a look at the following image, which shows the predicted and ground truth bounding boxes.



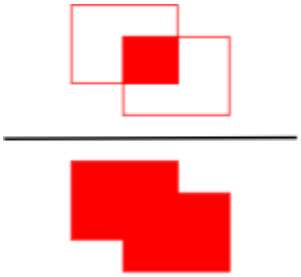
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Recall that this is similar to the actual and predicted outputs, which you have been comparing in the case of ML and deep learning (DL) algorithms so far. However, instead of having a single value of the outputs, in the case of object detection algorithms, you will have a vector.

The IoU is measured by taking the intersection of the two boxes and dividing it by their union. IoU can be expressed by the following formula:

IoU = Area of intersection / Area of union

The image given below depicts the numerator and the denominator of this formula.



Note that IoU lies between 0 and 1, since it is a fraction.

Next, you understood how to compute loss. Object detection algorithms have two losses, which are:

1. Classification or softmax loss, and
2. Regression losses – L1 and L2 losses.

You are already familiar with these losses, but let's revisit the L1 and L2 losses for object detection algorithms. **L1 loss** is employed to minimise the error that is defined as the sum of all the absolute deviations between the true and the projected values.

The formula for L1 loss is as follows:

$$L1LossFunction = \sum_{i=1}^{n} |y_{true} - y_{predicted}|$$

L2 loss is used to reduce the error that is defined as the total of all the squared discrepancies between the true and the predicted values. As you already know, L2 loss is the more preferred option in the case of outliers in data.

The formula for L2 loss is as follows:

$$L2LossFunction = \sum_{i=1}^{n} (y_{true} - y_{predicted})^2$$

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